4. Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.

Aim:

To implement a polyalphabetic substitution cipher using a keyword to encrypt the plaintext.

Procedure:

1. Read the plaintext and key.
2. Convert letters to uppercase.
3. Shift each plaintext letter using the corresponding key letter.
4. Repeat the key when necessary.
5. Display the ciphertext.

Program (C):

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
int main() {
    char text[100], key[100];
    int i, j = 0, k;
    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);
    printf("Enter key: ");
    scanf("%99s", key);
    for (i = 0; text[i] != '\0'; i++) {
        if (isalpha((unsigned char)text[i])) {
            k = toupper(key[j % strlen(key)]) - 'A';
            text[i] = (toupper(text[i]) - 'A' + k) % 26 + 'A';
```

```
            j++;
        }
    }
    printf("Ciphertext: %s", text);
    return 0;
}
```

Input:
Plaintext: HELLO
Key: KEY

Output:

```
Output

Enter plaintext: HELLO
Enter key: KEY
Ciphertext: RIJVS
```

Result:

Thus, the plaintext was encrypted successfully using the Vigenère cipher.